Router and Switch Configuration Project

Project Description

This project demonstrates the configuration of routers and switches using **Cisco Packet Tracer**

The goal of this project is to establish communication between two different LAN networks by assigning IP addresses and configuring router interfaces. Routers and switches are essential networking devices used in real-world IT infrastructures. This project provides hands-on experience in configuring network devices and verifying connectivity between networks.

Tools Used

The following tool was used to design and simulate the network:

- Cisco Packet Tracer

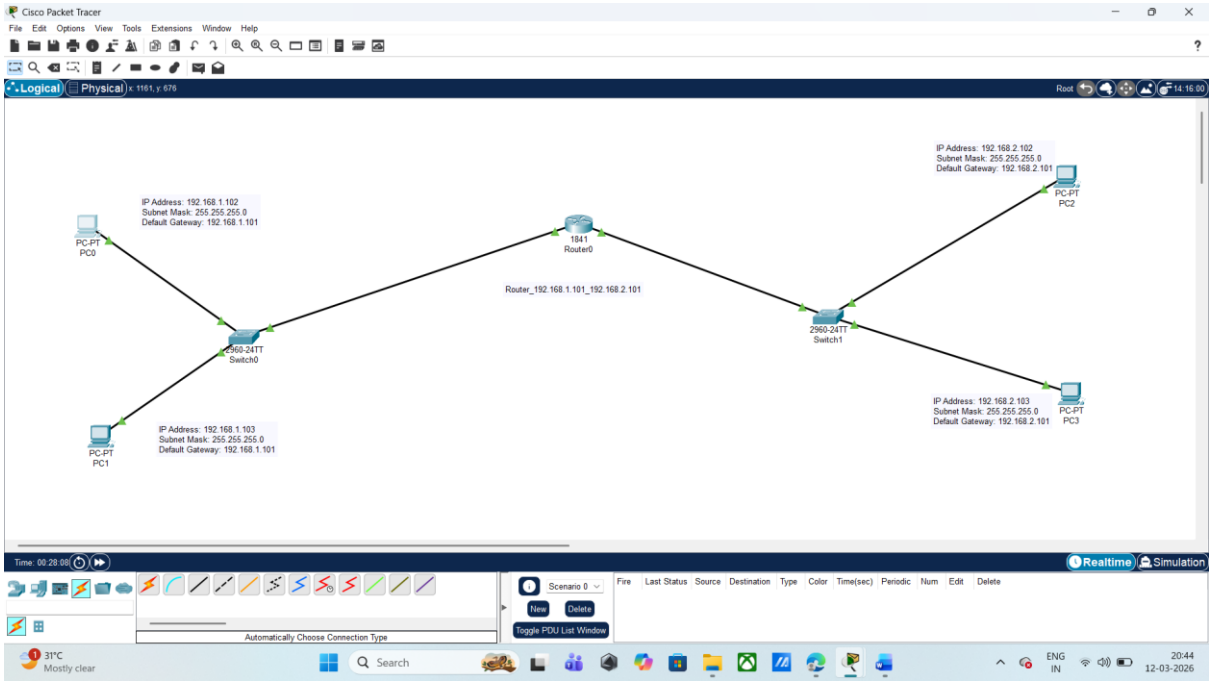
This tool allows users to create network topologies and configure networking devices such as routers, switches, and computers

NETWORK TOPOLOGY

The network consists of the following devices:

- 1 Router (Cisco 1841)
- 2 Switches (Cisco 2960)
- 4 PCs

Two LAN networks are connected through a router.
Network 1: 192.168.1.0 /24
Network 2: 192.168.2.0 /24
Each LAN is connected to a switch, and the switches are connected to the router to enable communication between both networks.

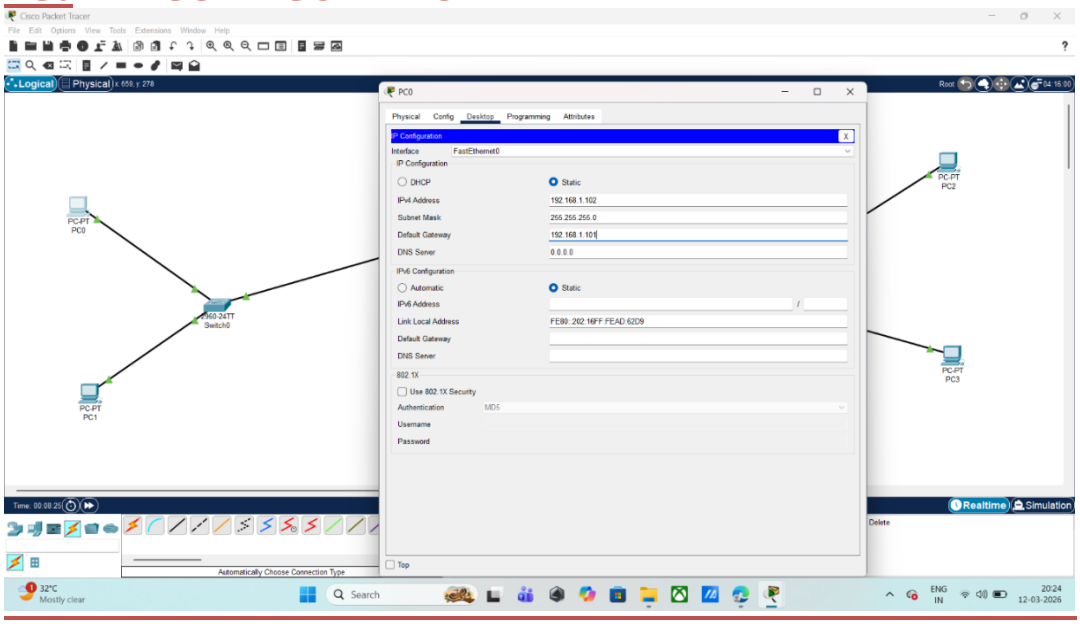


IP CONFIGURATION OF ALL PCS

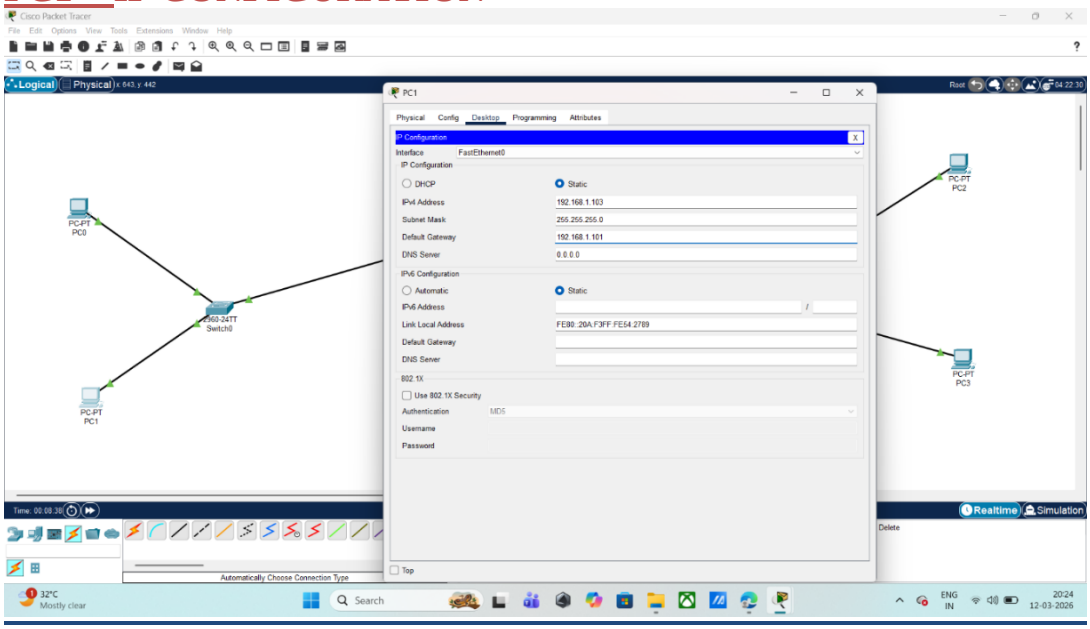
Each PC was assigned an IP address according to its respective network.

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.102	255.255.255.0	192.168.1.101
PC1	192.168.1.103	255.255.255.0	192.168.1.101
PC2	192.168.2.102	255.255.255.0	192.168.2.101
PC3	192.168.2.103	255.255.255.0	192.168.2.101

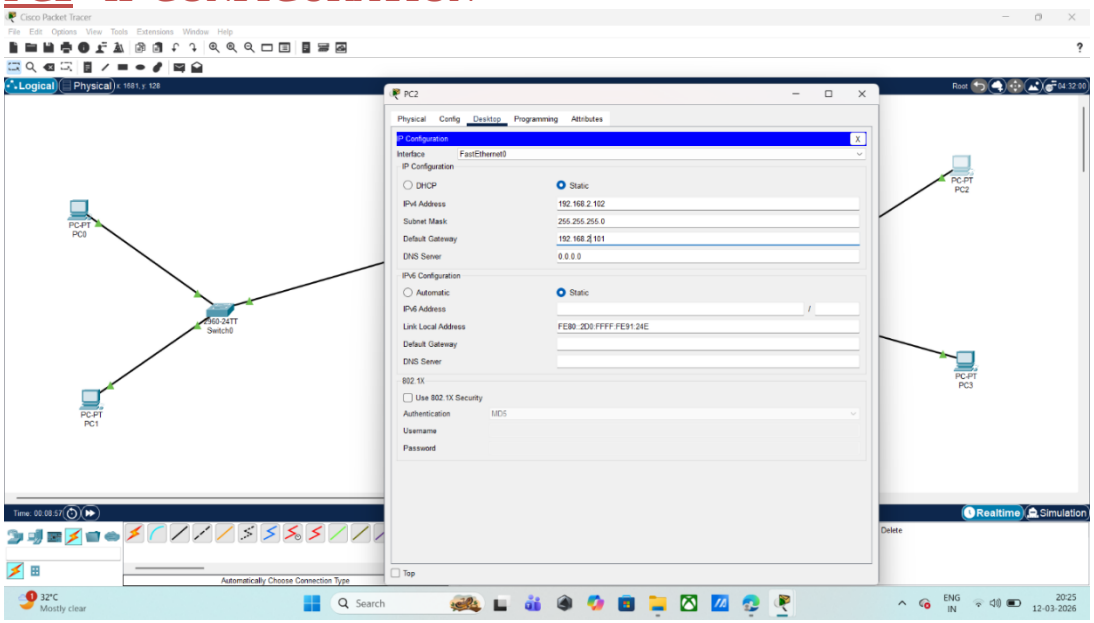
PC0 - IP CONFIGURATION



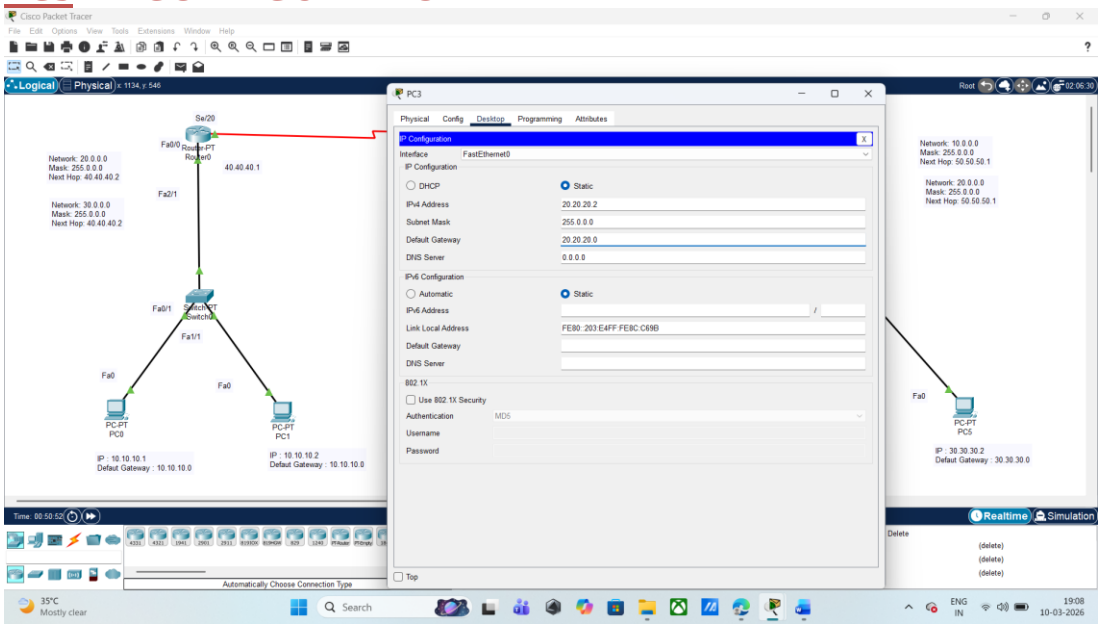
PC1 - IP CONFIGURATION



PC2 - IP CONFIGURATION



PC3- IP CONFIGURATION



ROUTER CONFIGURATION

The router was configured to connect both LAN networks using two FastEthernet interfaces.

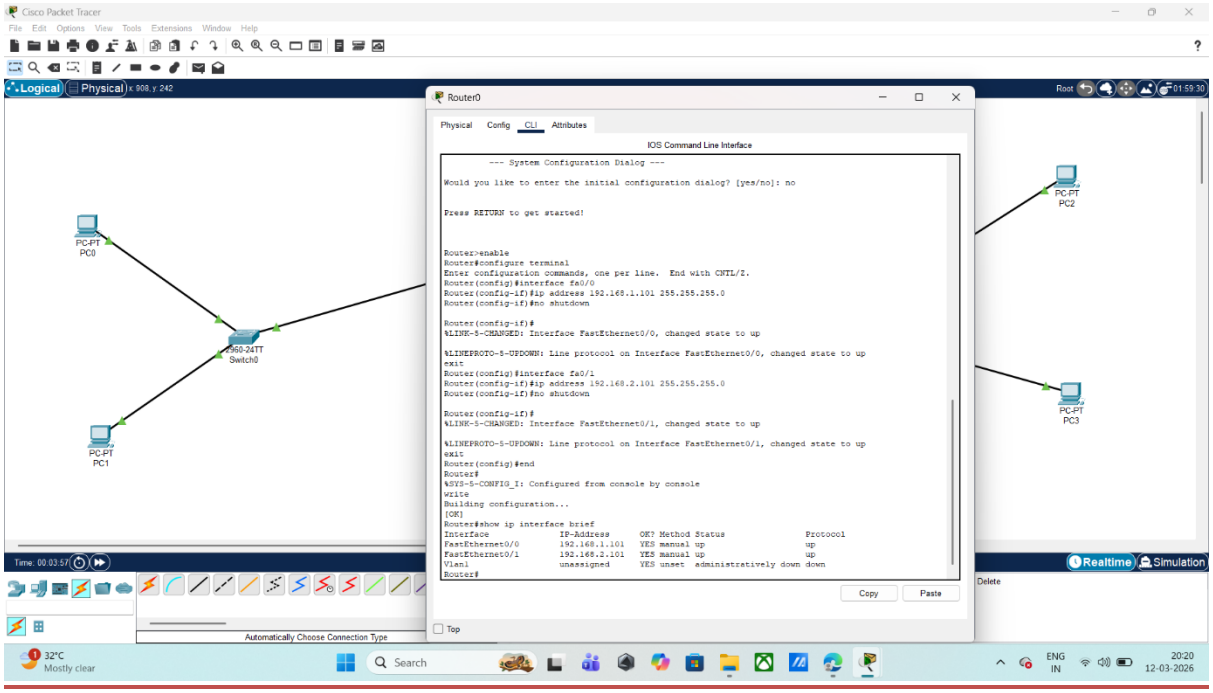
- FastEthernet0/0 → Network 192.168.1.0
- FastEthernet0/1 → Network 192.168.2.

Router configuration commands:

enable
configure terminal

interface fastethernet0/0
ip address 192.168.1.101 255.255.255.0
no shutdown

interface fastethernet0/1
ip address 192.168.2.101 255.255.255.0
no shutdown



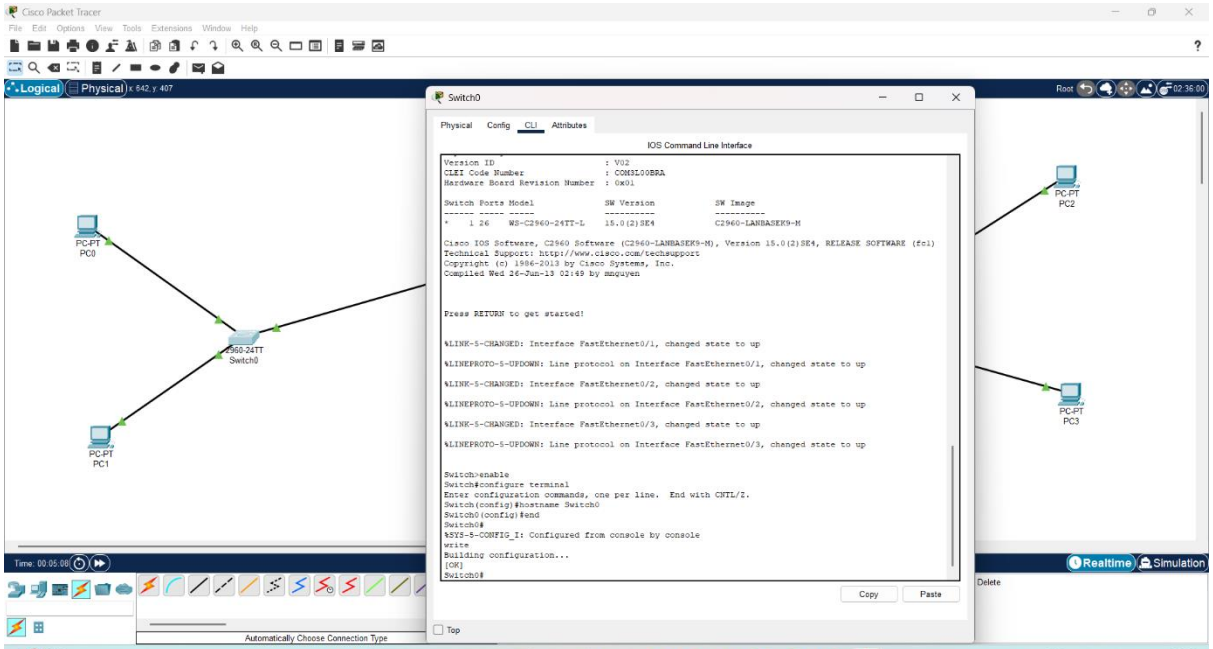
SWITCH CONFIGURATION

Switches are used to connect multiple PCs within each LAN network.
Basic configuration includes assigning hostnames to identify each switch.

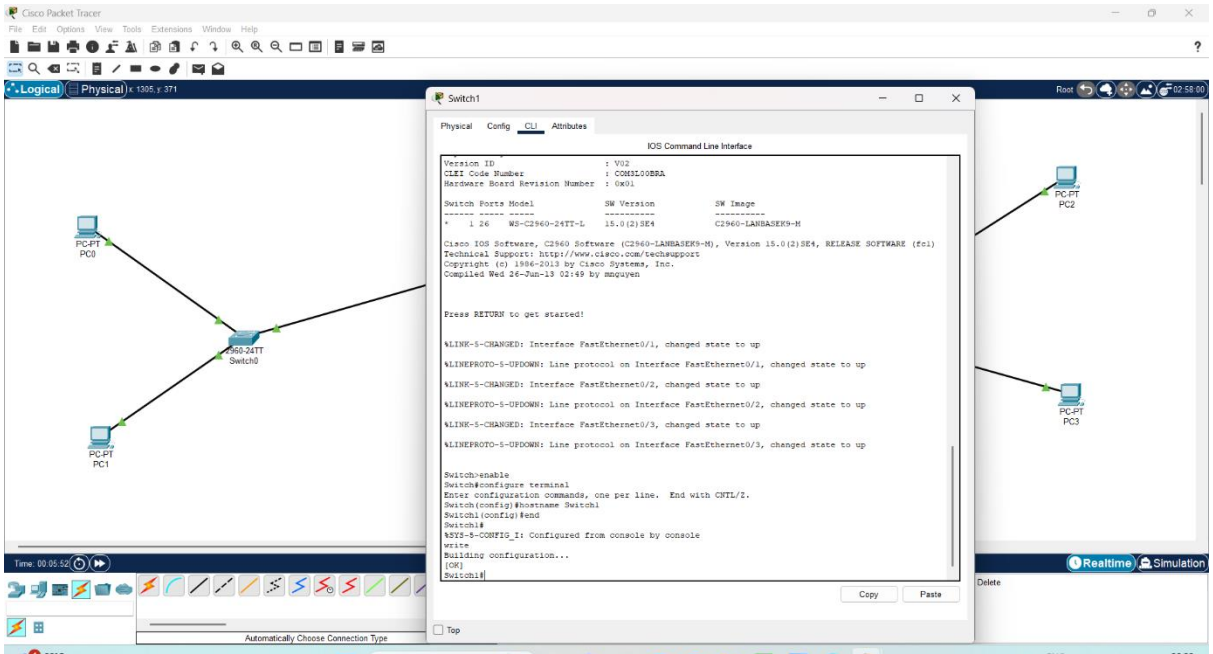
Example configuration:

enable
configure terminal
hostname Switch0
The same process was followed for the second switch.

SWITCH 0 CONFIGURATION



SWITCH 1 CONFIGURATION



TEST NETWORK

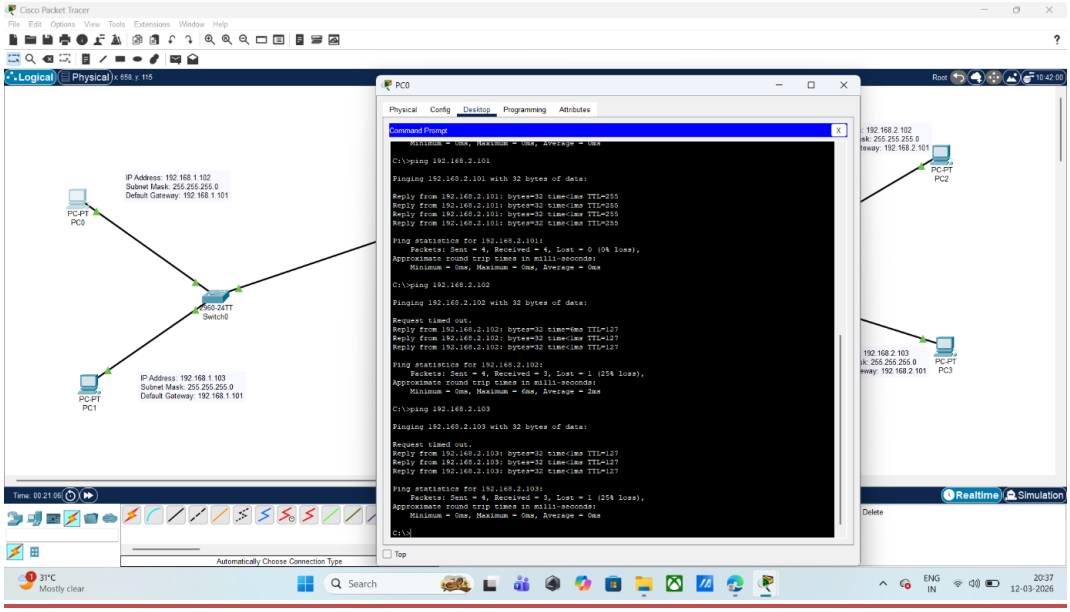
After completing the configuration, network connectivity was tested using the ping command.

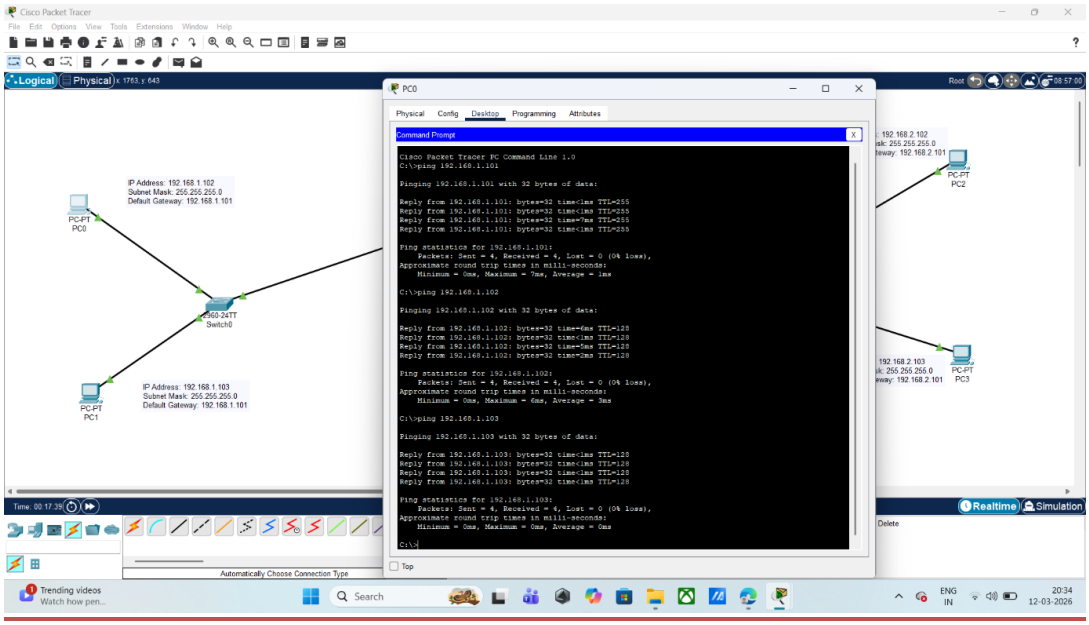
Example commands used:

ping 192.168.2.102

ping 192.168.1.102

Successful replies confirm that the router correctly routes packets between the two networks.





RESULT

The router and switches were successfully configured. All PCs in both LAN networks were able to communicate with each other. This confirms that routing between the two networks was successfully implemented.

CONCLUSION

This project demonstrates the configuration of routers and switches using Cisco Packet Tracer. It provides practical knowledge of IP addressing, LAN communication, and routing between different networks.
The successful ping tests confirm that communication between the two LAN networks was achieved through proper configuration of networking devices.